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Mars Telecommunications Network

Mars Telecommunications Network (MTN) Objectives and Requirements Document



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Table of Contents

1	INTRODUCTION	1
1.1	Purpose.....	1
1.2	Scope.....	1
1.3	Applicability	1
1.4	Stakeholders	1
2	REFERENCE AND APPLICABLE DOCUMENTS	2
2.1	Applicable Documents.....	2
2.2	Reference Documents	2
3	GOALS AND OBJECTIVES	3
4	GROUND RULES AND ASSUMPTIONS	3
5	MARS TELECOMMUNICAITONS NETWORK REQUIREMENTS	4
5.1	Project Requirements	4
5.2	Vehicle Requirements	8
5.3	Direct with Earth (DWE) Link Requirements	11
5.4	Proximity Link Requirements.....	14
	Appendix A Abbreviations and Acronyms	20

List of Tables

Table 1 Ground Rules and Assumptions	4
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PRELIMINARY

1 INTRODUCTION

Consistent with law, the National Aeronautics and Space Administration (NASA) is working toward a high-performance Mars telecommunications orbiter(s), capable of providing assets in and around Mars with robust, continuous communications with earth.

1.1 Purpose

The purpose of this Objectives and Requirements Document is to provide objectives and requirements for the Mars Telecommunications Network Project.

The intent of these objectives and requirements is to support definition of the required “what” and to the maximum extent possible not define the “how”. However, since existing assets are already operating in and around Mars, legacy compatibility is required, and as such detailed specificity is provided in this document and associated applicable and reference documents. The longer-term vision is evolution of the existing Mars network to a higher performing system with extensibility to a 24/7 end state.

1.2 Scope

The scope of this requirements document includes orbiter(s), direct with earth (DWE), and proximity link related requirements. The document does not include specific launch vehicle-related requirements. This document also describes top-level objectives of the MTN orbiter(s).

1.3 Applicability

This objectives and requirements document is applicable to the Mars Telecommunications Network Project.

1.4 Stakeholders

The following are the key stakeholders for the MTN Project:

- Exploration Systems Development Mission Directorate (ESDMD)
- Space Operations Mission Directorate (SOMD)
- Strategy and Architecture Office (SAO)
- Science Mission Directorate (SMD)
- Space Communication and Navigation (SCaN) Program
- Moon to Mars (M2M)

2 REFERENCE AND APPLICABLE DOCUMENTS

2.1 Applicable Documents

Applicable documents are those documents that form a part of this document. These documents carry the same weight as if they were stated within the body of this document.

Document Number	Document Title	Revision
<i>Architecture/Mission</i>		
NASA STD-871927	Implementing Planetary Protection Requirements for Spaceflight	
JPL D-106417	Mars Handbook 2022-2040	1.0
<i>Direct with Earth Links</i>		
810-007	DSN Mission Interface Design Handbook	
810-005	DSN Telecommunications Link Design Handbook	
503.0-B-2	CCSDS Tracking Data Message	
502.0-B-2	CCSDS Orbit Data Message	
<i>Proximity Links</i>		
	MRN Participation Guide	
	MRN Participation Guide Technical Support Package	
	CCSDS Bundle Protocol Specification	
	CCSDS Prox-1 Space Link Protocol - Data Link Layer	
732.0-B-4	CCSDS AOS Space Data Link Protocol	
732.1-B-3	CCSDS Unified Space Data Link Protocol	
	CCSDS Prox-1 Space Link Protocol - Coding and Synchronization Sublayer	
	CCSDS Prox-1 Space Link Protocol - Physical Layer	
TBD-01	Legacy Proximity Link Implementation (PICS)	
<i>NASA NPR's and Standards</i>		
NASA-STD-7009	Standard for Models and Simulations	B

2.2 Reference Documents

Reference documents are those documents that, though not a part of this document, serve to clarify the intent and contents of this document.

Document Number	Document Title	Revision
<i>Architecture/Mission</i>		
	M2M Architecture Description Document	C
NIST.FIPS.140-3	Federal Information Processing Standards Publication	
	SCaN Architecture Description Document	
	Planetary Protection Handbook 2024	
<i>Direct with Earth Links</i>		
104-DSN	34-m BWG Stations Telecommunications Interfaces	

875-0001	DSN Mission Service Interfaces, Policies, and Practices (MSIPP)	H
820-10	Deep Space Network Services Catalog	H
NASA NPR's and Standards		
NPR 7123.001	NASA System Engineering Processes and Requirements	D

3 GOALS AND OBJECTIVES

The top-level objectives of the MTN project are identified as:

MTN-OBJ-001

Provide communication and data exchange between assets in the Mars vicinity, the Mars surface, and Earth anticipated to operate at Mars through 2035.

MTN-OBJ-002

Provide Doppler, range, and time transfer to support positioning, navigation, and timing for assets anticipated to operate at Mars through 2035.

MTN-OBJ-003

Provide communication services to existing operational missions.

MTN-OBJ-004

Provide communications, Doppler, range, and time transfer services to the Entry, Descent, and Landing (EDL) demonstration missions anticipated to operate at Mars through 2035.

4 GROUND RULES AND ASSUMPTIONS

The ground rules and assumptions are identified and defined in the following table:

ID	Ground Rule / Assumption
1	MTN will not require surface-to-surface relay communications for 2035 uses (not precluded either).
2	MTN will not require orbiter crosslink capability (extensibility) (not precluded either). 24/7 end state extensibility is envisioned in the future.
3	DSN compatibility is required.
4	Radio to radio ground compatibility testing is required.
5	Contractor will provide portable test unit for proximity link compatibility testing.
6	MTN will require adaptive data rates on the proximity link, consistent with existing capability.
7	MTN will not require optical for DWE link (not precluded either).
8	MTN will require high rate DWE link and DWE connectivity throughout all operational phases.

9	MTN will require FWD and RTN Legacy and Advanced Proximity links.
10	MTN will require UHF FWD and RTN Legacy Proximity link.
11	MTN is procuring a system via FFP, and scope includes delivery of ground and flight hardware, not necessarily launch. Launch and future operations will be handled via contract options.
12	No specific requirements are included for a particular launch vehicle interface, that will be managed by the contractor.
13	MTN will require scheduling functionality and associated interfaces for both DWE and Proximity links
14	SMD payload is not precluded, and schedule risk is critically important.

Table 1 Ground Rules and Assumptions

5 MARS TELECOMMUNICAIONS NETWORK REQUIREMENTS

5.1 Project Requirements

MTN-REQ-001 *Operational Lifetime*

Threshold Requirement: Mars Telecommunications Network shall provide communications and Position, Navigation, and Timing (PNT) services for a minimum operational lifetime of 5 years, beginning at operational readiness.

Goal Requirement: N/A

Rationale: A 5-year operational lifetime ensures the Mars Telecommunications Network architecture can provide stable relay services across multiple Mars missions and demonstrations beginning at operational readiness. For the purposes of this requirement, operational readiness is defined as post-commissioning and ready to provide full operational capability. (Ref: Moon-to-Mars Architecture Description Document Rev C)

Verification Method: Analysis

Traceability: MTN-OBJ-001, MTN-OBJ-002, MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-002 *Proximity Link Continuity of Service*

Threshold Requirement: Mars Telecommunications Network shall provide Proximity Link radiometric and communications services to current and in development Mars surface assets.

Goal Requirement: N/A

Rationale: Integrated communications and radiometric services are required to support commanding, telemetry, and precision navigation for existing and in-development Mars surface assets.

Verification Method: Test

Traceability: MTN-OBJ-001, MTN-OBJ-002, MTN-OBJ-003

MTN-REQ-003 *Non-Real-Time Store-and-Forward*

Threshold Requirement: Mars Telecommunications Network shall provide store-and-forward capability for Direct with Earth (DWE) and proximity communications

Goal Requirement: N/A

Rationale: Providing non-real-time, store-and-forward enables reliable data delivery despite long delays, intermittent visibility, and periodic loss of line-of-sight between surface, orbital, and Earth segments ensuring continuous mission communications in Mars orbital geometries without requiring real-time connectivity.

Verification Method: Test

Traceability: MTN-OBJ-001, MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-004 *Concurrent Service Provision*

Threshold Requirement: Mars Telecommunications Network shall support concurrent full duplex operation of communications relay DWE, Proximity communications, and associated services.

Goal Requirement: N/A

Rationale: Concurrent, non-interfering operation of communications relay, semi-real-time delivery, DTN networking, navigation, timing, and messaging is required to support overlapping mission activities without degrading critical services. This capability ensures predictable performance under peak load, protects time-critical functions, and enables Mars Telecommunications Network to operate as reliable, multi-mission infrastructure consistent with SCan architecture and NASA risk-informed systems engineering practices (SCaN ADD, CCSDS BPv7, NPR 7123.1, NPR 8705.5).

Verification Method: Test

Traceability: MTN-OBJ-001, MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-005 *Hardware Delivery Date*

Threshold Requirement: The contractor shall deliver completed hardware delivery to NASA including completion of all associated reviews and inspections by December 31, 2028.

Goal Requirement: N/A

Rationale: Consistent with Pub. L. 119-21, delivery by end of CY2028 ensures continuity of critical relay services and prevents a communications gap as existing Mars orbiters age.

Verification Method: Inspection

Traceability: MTN-OBJ-001, MTN-OBJ-002, MTN-OBJ-003

MTN-REQ-006 *Direct with Earth (DWE) Service*

Threshold Requirement: Mars Telecommunications Relay shall provide Direct with Earth (DWE) under all flight phases.

Goal Requirement: N/A

Rationale: The Mars Telecommunications Network will provide reliable, high-rate communication and radiometric services with terrestrial ground-stations in support of current, in-development and future Mars efforts.

Verification Method: Test

Traceability: MTN-OBJ-001, MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-007 *Mission Environment*

Threshold Requirement: Mars Telecommunications Relay shall operate as specified herein when exposed to all expected environments through all expected mission phases.

Goal Requirement: N/A

Rationale: The Mars Telecommunications Network, in addition to on-orbit lifecycle, will be exposed to all relevant operational environments through deployment into orbit including but not limited to orbital insertion, time-light delays from ground command preventing timely intervention and potential intermittent loss of communication and command. [Ref. Mars Handbook 2022-2040 Release 1.0]

Verification Method: Analysis

Traceability: MTN-OBJ-001, MTN-OBJ-002, MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-008 *Solar Conjunction*

Threshold Requirement: The Mars Telecommunications Network Direct with Earth (DWE) connections shall operate at Sun-Earth-probe angles greater than 3°

Goal Requirement: The Mars Telecommunications Network Direct with Earth (DWE) connections shall operate at Sun-Earth-probe angle greater than 2°

Rationale: The Earth-Mars solar conjunction will prevent communications between Earth and Mars for the duration the time of the actual line-of-sight blockage. Designing communication links to operate within 3° of the blockage will increase the availability of the links by incorporating mitigation techniques and link margin to optimize communications in the presence of solar adverse effects on the respective receivers. Operational constraints may be imposed to mitigate the potential of corrupted commands sent to the spacecraft. For this requirement, threshold values reflect expected performance of X-Band transmission.

Verification Method: Analysis

Traceability: MTN-OBJ-001, MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-009 *Ground Commanded Prioritization*

Threshold Requirement: The Mars Telecommunications Network shall provide the capability for data prioritization to be set from the ground user under nominal conditions.

Goal Requirement: N/A

Rationale: The Mars Telecommunications Network will service multiple end users. Therefore, it is necessary to provide a means of prioritizing data to facilitate link scheduling.

Verification Method: Inspection

Traceability: MTN-OBJ-001, MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-010 *Scheduling Functionality*

Threshold Requirement: The Mars Telecommunications Network shall implement a means for user scheduling and providing associated interfaces for both Direct with Earth (DWE) and Proximity links.

Goal Requirement: N/A

Rationale: The Mars Telecommunications Network will service multiple end users. Therefore, it is necessary to provide for scheduling functionality and the associated interfaces for user data.

Verification Method: Inspection

Traceability: MTN-OBJ-001, MTN-OBJ-002, MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-011 *Link Performance*

Threshold Requirement: The Mars Telecommunications Network shall meet end-to-end performance for all links with ≥ 3 dB margin.

Goal Requirement: N/A

Rationale: Meeting end-to-end performance at maximum range with at least 3 dB margin ensures reliable communications under nominal. The ≥ 3 dB link margin requirement provides explicit robustness against modeling uncertainty, hardware performance drift, pointing error, aging of RF components, and unmodeled environmental effects. This aligns with NASA systems engineering guidance (e.g., NPR 7123.1 and NASA-STD-7009) to include appropriate performance margin for model-based predictions and long-life space assets, reducing the risk of link outages, increased retransmissions, or permanent loss of service.

Verification Method: Test

Traceability: MTN-OBJ-001, MTN-OBJ-002, MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-012 *Bit Error rate*

Threshold Requirement: All Mars Telecommunications Network links shall have a bit error rate of 10^{-7} or better.

Goal Requirement: N/A

Rationale: This value is aligned with NASA best practice and ensures for accurate decryption of command link.

Verification Method: Analysis

Traceability: MTN-OBJ-001, MTN-OBJ-003, MTN-OBJ-004

5.2 Vehicle Requirements

MTN-REQ-013 *Operational Readiness*

Threshold Requirement: Mars Telecommunications Network shall achieve full operational readiness no later than end of calendar year 2030.

Goal Requirement: Mars Telecommunications Network shall achieve full operational readiness no later than end of calendar year 2029.

Rationale: Achieving Mars Telecommunications Network operational readiness by the end of 2030 ensures continuity of critical relay services and prevents a communications gap as existing Mars orbiters age. Aligning Mars Telecommunications Network's operational readiness with the 2028–2029 Mars launch opportunity reduces risk to dependent surface and orbital missions and supports stable programmatic planning and sustained Mars exploration. This is consistent with Pub. L. 119-21, to deliver to NASA by end of CY20208. For the purposes of this requirement,

operational readiness is defined as post-commissioning and ready to provide full operational capability.

Verification Method: Analysis

Traceability: MTN-OBJ-001, MTN-OBJ-002, MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-014 *Non-Volatile Store-and-Forward Memory (DWE)*

Threshold Requirement: Mars Telecommunications Network shall provide at least 1.0 Tbyte of non-volatile store and forward capability for the DWE link.

Goal Requirement: N/A

Rationale: Providing non-real-time, store-and-forward enables reliable data delivery despite long delays, intermittent visibility, and periodic loss of line-of-sight between surface, orbital, and Earth segments ensuring reliable mission communications in Mars orbital geometries without requiring real-time connectivity. This value reflects an assumed average data-rate of 7 Mbps over an 8-hour pass per day for 2 days with a factor of 10 margin.

Verification Method: Inspection

Traceability: MTN-OBJ-001, MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-015 *Non-Volatile Store-and-Forward Memory (Proximity)*

Threshold Requirement: Mars Telecommunications Network shall provide at least 0.5 Tbyte of non-volatile store and forward capability for the Proximity link(s).

Goal Requirement: N/A

Rationale: Providing non-real-time, store-and-forward enables reliable data delivery despite long delays, intermittent visibility, and periodic loss of line-of-sight between surface, orbital, and Earth segments ensuring reliable mission communications in Mars orbital geometries without requiring real-time connectivity.

Verification Method: Inspection

Traceability: MTN-OBJ-001, MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-016 *Cybersecurity & Command Protection*

Threshold Requirement: Mars Telecommunications Network shall utilize cybersecurity and command link encryption practices consistent with industry best practices.

Goal Requirement: N/A

Rationale: NASA-approved security is required to protect Mars Telecommunications Network commanding, data integrity, and authentication against unauthorized access and data compromise. It is encouraged to implement existing industry practices where appropriate.

Verification Method: Demonstration

Traceability: MTN-OBJ-001, MTN-OBJ-002, MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-017 *Planetary Protection Guidelines*

Threshold Requirement: Mars Telecommunications Network shall follow all Agency applicable Planetary Protection guidelines (NPD 1000.3, NASA/SP-20240016475, NASA-STD-8719.27).
Goal Requirement: N/A

Rationale: As a Class 3 Mars Orbital mission per NPR 8719.27, Mars Telecommunications Network is required to provide NASA access to the spacecraft, facilities, and documentation to accommodate planned independent verification and assurance activities, development of risk mitigation protocols, and any anomaly investigations throughout the project life cycle to achieve compliance with applicable Agency planetary protection requirements.

Verification Method: Inspection

Traceability: MTN-OBJ-001, MTN-OBJ-002, MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-018 *Orbital Debris*

Threshold Requirement: Mars Telecommunications Network shall follow all Agency applicable Mars Orbital Debris requirements (NASA-STD-8719.14, NPR 8715.6).

Goal Requirement: N/A

Rationale: Mars Telecommunications Network is required to demonstrate compliance with NASA Mars Orbital debris reduction requirements NASA-STD-8719.14 and NPR 8715.6 to limit debris generation, spacecraft vulnerability, post-mission lifetime, and maximize entry safety.

Verification Method: Analysis

Traceability: MTN-OBJ-001, MTN-OBJ-002, MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-019 *Maximum Expected DWE Data Rate*

Threshold Requirement: The Mars Telecommunications Network shall have onboard capability to support data rates of up to 100 Mbps on the Direct with Earth links.

Goal Requirement: N/A

Rationale: The Mars Telecommunications Network will be capable of supporting high-rate communications to the maximum rates expected under favorable conditions and close proximity with Earth to prevent hardware bottlenecks.

Verification Method: Test

Traceability: MTN-OBJ-001, MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-020 *Minimum Orbit Altitude*

Threshold Requirement: The Mars Telecommunications Network shall have a minimum orbit altitude around Mars of at least 600 km.

Goal Requirement: N/A

Rationale: The Mars Telecommunications Network shall provide service to orbital assets up to an altitude of 500km. This requirement in conjunction with the Proximity Link Orbital Coverage requirements satisfies this function.

Verification Method: Analysis

Traceability: MTN-OBJ-001, MTN-OBJ-002, MTN-OBJ-003, MTN-OBJ-004

5.3 Direct with Earth (DWE) Link Requirements

MTN-REQ-021 *DWE Link 1-Way Doppler / Pseudo-range (PR) Reference Signals*

Threshold Requirement: Mars Telecommunications Network shall be able to collect pseudo-range (PR) and 1-way forward Doppler on signals from the DWE links.

Goal Requirement: N/A

Rationale: Pseudo-range and one-way forward Doppler reference signals support navigation when two-way tracking is unavailable or impractical. These services support high-dynamics mission phases, reduce onboard complexity for users, and improve navigation continuity, autonomy, and overall mission resilience.

Verification Method: Test

Traceability: MTN-OBJ-002

MTN-REQ-022 *DWE Link 2-Way Coherent Ranging & Doppler*

Threshold Requirement: Mars Telecommunications Network shall be able to support 2-way coherent ranging and 2-way coherent Doppler on the DWE links with a precision of range precision of 1 m and Doppler precision of .1 mm/s at a 60 second count time.

Goal Requirement: N/A

Rationale: Two-way coherent ranging and Doppler provide the highest-accuracy range and velocity measurements needed for precise navigation and orbit determination. These capabilities are essential for high-consequence operations and reduce mission risk by enabling stable, validated navigation solutions. Precision replicates current accuracy capability of existing systems and reflects 1-sigma at X-Band.

Verification Method: Test

Traceability: MTN-OBJ-002

MTN-REQ-023 *DWE Link Communication Availability*

Threshold Requirement: Mars Telecommunications Network shall provide an average annual DWE link service availability of $\geq 95\%$, exclusive of the Earth side DWE availability.

Goal Requirement: Mars Telecommunications Network shall provide an average annual DWE link service availability of $\geq 99\%$, exclusive of the Earth side DWE availability.

Rationale: A service availability of at least 95% ensures Mars Telecommunications Network provides reliable, predictable support for mission-critical communications and navigation. This requirement drives resilient system design and reduces operational risk for sustained Mars exploration. This is achieved through a combination of low and high rate DWE communications. This requirement is on the Mars Telecommunications Network portion of the link and not the availability of the Earth side of the link.

Verification Method: Analysis

Traceability: MTN-OBJ-001, MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-024 *DWE Link Data Rate*

Threshold Requirement: Mars Telecommunications Network shall support DWE Link data rates ≥ 0.1 Mbps Forward and ≥ 1 Mbps Return at Mars–Earth maximum range (~400Mkm).

Goal Requirement: Mars Telecommunications Network shall support DWE Link data rates ≥ 1 Mbps Forward and ≥ 7 Mbps Return at Mars–Earth maximum range (~400Mkm).

Rationale: Specifying minimum DWE data rates at Mars–Earth maximum range ensures Mars Telecommunications Network can provide reliable commanding and high-volume data return under worst-case link conditions. Specified rates support efficient, high-throughput operations consistent with DSN services. Designing to these thresholds drives DSN compatibility and

supports resilient, campaign-level Mars Telecommunications Network infrastructure in accordance with DSN interface guidance, SCA architectural direction, and NASA systems engineering practices (DSN MIDH 810-007; DSN IRD; SCA ADD; NPR 7123.1; Mars Telecommunications Network Handbook).

Verification Method: Test

Traceability: MTN-OBJ-001, MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-025 *DWE Link / Deep Space Network (DSN) Compatibility*

Threshold Requirement: Mars Telecommunications Network shall be compatible with DSN per DSN Mission Interface Design Handbook (810-007) and DSN Telecommunications Link Design Handbook (810-005).

Goal Requirement: Mars Telecommunications Network shall be compatible with DSN per DSN Mission Interface Design Handbook (810-007), DSN Telecommunications Link Design Handbook (810-005) and provide functionality to support Multiple Users per Aperture (MUPA)

Rationale: Compatibility with the Deep Space Network (DSN) ensures that Mars Telecommunications Network can use NASA's primary interplanetary communications infrastructure without custom interfaces or redesign. The DSN Mission Interface Design Handbook (810-007) is the authoritative source for all flight projects using the DSN and provides required interface guidance, service capabilities, and customer requirements. Designing to 810-007 reduces technical and schedule risk, ensures interoperable communications and tracking services, and enables clear verification against an approved DSN baseline. Multiple Users per Aperture (MUPA) is a desired functionality which will require the ability to acquire an uplink signal when multiple spacecraft are using a single aperture.

Verification Method: Test

Traceability: MTN-OBJ-001, MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-026 *DWE Link / CCSDS Protocol Compatibility*

Threshold Requirement: Mars Telecommunications Network shall implement CCSDS BPv7 compliant communications interfaces and protocols.

Goal Requirement: N/A

Rationale: Implementing CCSDS BPv7-compliant interfaces ensures Mars Telecommunications Network can reliably interoperate with NASA, international, and commercial assets using standardized communications protocols.

Verification Method: Test

Traceability: MTN-OBJ-001, MTN-OBJ-003, MTN-OBJ-004

5.4 Proximity Link Requirements

MTN-REQ-027 *Proximity Link Orbital Coverage*

Threshold Requirement: Mars Telecommunications Network shall provide Proximity Link radiometric and communications services to future Mars orbital assets at orbital altitudes $\leq$ (Mars Telecommunications Network Altitude) - (100 km).

Goal Requirement: N/A

Rationale: The Mars Telecommunications Relay will provide proximity connection to orbital assets at orbital altitudes lower than that of the Mars Telecommunication Relay. This requirement provides clarity that support of assets at higher altitudes is not expected as a minimum.

Verification Method: Analysis

Traceability: MTN-OBJ-001, MTN-OBJ-002, MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-028 *Proximity Link Surface Coverage*

Threshold Requirement: Mars Telecommunications Network shall provide services for all longitudes and all latitudes of at least up to $\pm 50^\circ$ coverage of the Martian surface.

Goal Requirement: N/A

Rationale: Near-global coverage ensures the Mars Telecommunications Network system can provide reliable services across all longitudes and latitudes up to approximately $\pm 50^\circ$ on the Martian surface, with only limited polar exclusions. This capability supports current NASA missions and reduces operational risk by providing consistent, predictable relay access aligned with sustained, campaign-level Mars exploration objectives.

Verification Method: Analysis

Traceability: MTN-OBJ-001, MTN-OBJ-002, MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-029 *Proximity Link Time Transfer*

Threshold Requirement: Mars Telecommunications Network shall provide radiometric time-transfer services to surface and orbital assets with time-transfer accuracy $\leq$ 1 millisecond error relative to Coordinated Universal Time (UTC) at the user interface.

Goal Requirement: Mars Telecommunications Network shall provide radiometric time-transfer services to surface and orbital assets with time-transfer accuracy ≤ 1 microsecond error relative to Coordinated Universal Time (UTC) at the user interface.

Rationale: Accurate radiometric time-transfer provides a common time reference needed for precise positioning, navigation, and timing at Mars. This capability reduces navigation errors, supports time-critical operations, and enables reliable, interoperable PNT services for surface and orbital assets.

Verification Method: Test

Traceability: MTN-OBJ-002, MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-030 *Proximity Link Communication Availability*

Threshold Requirement: Mars Telecommunications Network shall provide an average annual Proximity Link service availability $\geq 98\%$ during the Proximity Communications windows.

Goal Requirement: Mars Telecommunications Network shall provide an average annual Proximity Link service availability $\geq 99\%$ during the Proximity Communications windows.

Rationale: A service availability of at least 98% ensures Mars Telecommunications Network provides reliable, predictable support for mission-critical communications and navigation. This requirement drives resilient system design and reduces operational risk for sustained Mars exploration.

Verification Method: Analysis

Traceability: MTN-OBJ-001, MTN-OBJ-002, MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-031 *UHF Proximity Link Data Rates*

Threshold Requirement: Mars Telecommunications Network shall support UHF Proximity Link communications services for surface and orbital assets at Mars with per-user data rates ≥ 2 Mbps under nominal proximity rate conditions at zenith.

Goal Requirement: Mars Telecommunications Network shall support UHF Proximity Link communications services for surface and orbital assets at Mars with per-user data rates ≥ 8 Mbps under nominal proximity rate conditions at zenith.

Rationale: Mars surface and orbital missions depend on proximity relay communications to return science, engineering, and health data when direct-to-Earth links are unavailable. Existing NASA Mars missions and relay orbiters establish a proven operational baseline for UHF proximity return-link throughput that the Mars Telecommunications Relay seeks to build upon. Specifying minimum proximity return-link data rates ensures continuity and interoperability with

deployed Mars assets while preserving flexibility in proximity relay architecture and implementation.

Verification Method: Test

Traceability: MTN-OBJ-001, MTN-OBJ-002, MTN-OBJ-003

MTN-REQ-032 *Advanced Proximity Link Data Rates*

Threshold Requirement: Mars Telecommunications Network shall support Advanced (non-UHF) Proximity Link communications for surface and orbital assets at Mars with per-user data rates ≥ 25 Mbps under nominal proximity rate conditions at zenith.

Goal Requirement: Mars Telecommunications Network shall support Advanced (non-UHF) Proximity Link communications for surface and orbital assets at Mars with per-user data rates ≥ 100 Mbps under nominal proximity rate conditions at zenith.

Rationale: Mars surface and orbital missions depend on proximity relay communications to return science, engineering, and health data when direct-to-Earth links are unavailable. Advanced proximity link data rates are derived from stakeholder input and anticipated needs within the MNT operational lifetime.

Verification Method: Test

Traceability: MTN-OBJ-001, MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-033 *Simultaneous Proximity Links*

Threshold Requirement: Mars Telecommunications Network shall simultaneously support ≥ 2 simultaneous concurrent full-duplex legacy links and ≥ 1 advanced proximity links within the Mars Telecommunications Network Proximity antenna fields of view.

Goal Requirement: N/A

Rationale: Supporting at least two concurrent full-duplex links enables Mars Telecommunications Network to provide simultaneous two-way communications for multiple surface and orbital users, reducing scheduling conflicts and supporting time-critical operations. This capability ensures Mars Telecommunications Network can function as shared, resilient campaign infrastructure with acceptable operational risk (NASA-HDBK-1002).

Verification Method: Test

Traceability: MTN-OBJ-001, MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-034 *Adaptive Data Rates*

Threshold Requirement: Proximity link system shall support adaptive data rates.

Goal Requirement: N/A

Rationale: Adaptive proximity links will achieve significant improvement in throughput capacity by using available link margin. Requires basic changes to the transmitter and/or receiver parameters including symbol rates, frequency channels, symbol coding, and others that affect the throughput and performance of the communications channels when conditions change.

Verification Method: Analysis

Traceability: MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-035 *Proximity Link 1-Way Doppler*

Threshold Requirement: The Mars Telecommunications Network shall provide for in situ 1-way Doppler tracking services to users with a precision of .5mm/s at a 60 second count time.

Goal Requirement: The Mars Telecommunications Network shall provide for in situ 1-way Doppler & Ranging services to users with a precision of .5mm/s at a 60 second count time (1-sigma, UHF) and ranging precision of 1 meter.

Rationale: Pseudo-range and one-way forward Doppler reference signals support navigation when two-way tracking is unavailable or impractical. These services support high-dynamics mission phases, reduce onboard complexity for users, and improve navigation continuity, autonomy, and overall mission resilience.

Verification Method: Test

Traceability: MTN-OBJ-002

MTN-REQ-036 *Proximity Link 2-Way Doppler*

Threshold Requirement: The Mars Telecommunications Network shall provide for in situ 2-Way coherent Doppler tracking services to users with a precision of .5mm/s at a 60 second count time.

Goal Requirement: The Mars Telecommunications Network shall provide for in situ 2-way coherent Doppler & Ranging services to users with a precision of .5mm/s at a 60 second count time and ranging precision of 1 meter.

Rationale: Two-way coherent ranging and Doppler provide the highest-accuracy range and velocity measurements needed for precise navigation and orbit determination. These capabilities are essential for high-consequence operations and reduce mission risk by enabling stable,

validated navigation solutions. Precision replicates current accuracy capability of existing systems.

Verification Method: Test

Traceability: MTN-OBJ-002

MTN-REQ-037 *Position, Navigation, and Timing (Service Reference Signals)*

Threshold Requirement: The Mars Telecommunications Network shall provide pseudorange (PR) and 1-way forward Doppler reference signals with a signal-in-space-errors (SISE) contributions of ≤ 100 meters

Goal Requirement: The Mars Telecommunications Network shall provide pseudorange (PR) and 1-way forward Doppler reference signals with a signal-in-space-errors (SISE) contributions of ≤ 50 meters

Rationale: The Mars Telecommunications Network shall be capable of providing reference signals that can be used by all assets at any time to calculate their own position and time. SISE includes errors from: relay on-board state knowledge (position, velocity and time), signal integrity and on-board path delays, predicted relay states conveyed to users, and states and time representation. SISE does not include delays and phase noise introduced by the user system. The threshold value reflects what is achievable utilizing procurable USO clocks. The Goal value assumes use of atomic clock system.

Verification Method: Test

Traceability: MTN-OBJ-002

MTN-REQ-038 *Proximity Link Protocol Standard*

Threshold Requirement: Mars Telecommunications network shall implement a CCSDS protocol standard.

Goal Requirement: N/A

Rationale: The Mars Telecommunications Network will implement a non-proprietary standard for proximity link protocol.

Verification Method: Test

Traceability: MTN-OBJ-001, MTN-OBJ-003, MTN-OBJ-004

MTN-REQ-039 *Open-Loop Recording*

Threshold Requirement: The Mars Telecommunications Network shall provide open-loop recording for critical events (10 minutes in duration) at a rate of 100 kHz at 8-10 bits per sample.

Goal Requirement: N/A

Rationale: The Mars Telecommunications Network shall support the needed functionality currently available to legacy users during critical events.

Verification Method: Test

Traceability: MTN-OBJ-003, MTN-OBJ-004

PRELIMINARY

Appendix A Abbreviations and Acronyms

ADD	Architecture Description Document
BP	Bundle Protocol
CCSDS	Consultative Committee for Space Data Systems
CY	Calendar Year
dB	Decibel
DSN	Deep Space Network
DWE	Direct with Earth
EDL	Entry, Descent, and Landing
ESDMD	Exploration Systems Development Mission
FWD	Forward
IRD	Interface Requirements Document
kHz	Kilohertz
km	Kilometer
m	meter
M2M	Moon to Mars
Mbps	Megabits per Second
mm	Millimeter
MTN	Mars Telecommunications Network
MUPA	Multiple Users Per Aperture
NASA	National Aeronautics and Space Administration
NPR	NASA Procedural Requirements
PNT	Position, Navigation, and Timing
RTN	Return
s	Second
SAO	Strategy and Architecture Office
SCaN	Space Communication and Navigation
SISE	Signal-in-Space-Error
SMD	Science Mission Directorate
SOMD	Space Operations Mission Directorate
Tbyte	Terabyte
UHF	Ultra-High Frequency
USO	Ultra-Stable Oscillator
UTC	Coordinated Universal Time